

# A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-123482119})$ , $p = 5$

Generated by `tools/certificate_guide.gp`

## What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value  $D_x(e_j)$  of a secondary norm operator: it stores the unramified cyclic quintic  $L_x/K$ , the normalized generator  $\sigma_x$ , the pair  $(a', J)$  representing the torsion class  $e_j$ , the auxiliary ideal  $I'$ , the compactly represented element  $t$ , a residue prime for the sign check, and the expected norm class  $[N_x(I')]$ . The two defining equations are

$$I'^{(1-\sigma_x)^2}(t) J \mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

## Notation and conventions

**Ideals as matrices.** An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a  $\mathbf{Z}$ -basis of the ideal. For example, the ideal  $J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}$  of Entry 1 below, over  $K$  with integral basis  $(1, s)$ , denotes the ideal  $\mathbf{Z} 2306 + \mathbf{Z} (1001 + s)$ . The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

**Which integral basis.** The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for  $K$ . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant  $\pm 1$  – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

**Characters.** The header fixes three generators  $\kappa_1, \kappa_2, \kappa_3$  of  $\text{Cl}(K)$  (as HNF ideals). A character  $x$  on  $\text{Cl}(K)/5$  is recorded by its value vector  $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$ ; thus  $(1, 0, 0)$  is the character dual to  $\bar{\kappa}_1$ . The column number  $j$  records which basis element  $e_j$  of  $\text{Cl}(K)[5]$  the stored pair  $(a', J)$  represents.

**Automorphisms.**  $\sigma_x$  is a field automorphism of  $L_x$ . Writing  $\theta$  for the chosen root of the stored absolute polynomial, so that  $L_x = \mathbf{Q}(\theta)$ , the automorphism is determined by the single value  $\sigma_x(\theta)$ , and the certificate stores the coordinate vector of  $\sigma_x(\theta)$  in the integral basis of  $L_x$ .

## Base field data

The base field is  $K = \mathbf{Q}(s)$  with  $s^2 - s + 30870530 = 0$ , of discriminant  $-123482119$ . Class group invariants  $(270 \ 5 \ 5)$ , class number 6750; the three stored generator ideals (HNF) are  $\begin{pmatrix} 3701 & 407 \\ 0 & 1 \end{pmatrix}$ ,  $\begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}$ ,  $\begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}$ , and the torsion units are generated by  $-1$  of order 2.

## Entry 1: character $a$ , column $e_1$

Prescribed character vector  $(1 \ 0 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$ .

## The pair $(a', J)$ representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

## The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $301379643833787 = 3^4 \cdot 3720736343627$ :

$$\begin{pmatrix} 11162209030881 & 9694892882580 & 9429338425302 & 10539739703127 & 728631965825 & 4983667736004 & 484741371976 & 341684204281 & 7940809015466 & 7691241836001 \\ 0 & 3 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 3 & 2 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

## Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (0 \ 4 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 2: character $a$ , column $e_2$

Prescribed character vector  $(1 \ 0 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $183134042912161 = 11 \cdot 12721 \cdot 1308745331$ :

$$\begin{pmatrix} 183134042912161 & 20800793765058 & 138973395532151 & 1815266478758 & 117730674209745 & 130437536825493 & 124815487559744 & 50948764649726 & 114660011809634 & 149276735157783 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 3$  with residue degree 2; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (0 \ 0 \ 3)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 3: character $a$ , column $e_3$

Prescribed character vector  $(1 \ 0 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$ .

## The pair $(a', J)$ representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

## The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $113043947459949 = 3^4 \cdot 11^3 \cdot 157 \cdot 331 \cdot 20177$ :

$$\begin{pmatrix} 103805277741 & 79528556151 & 60665060555 & 14789234891 & 95688952133 & 19692499249 & 45724492741 & 75503706402 & 16594326331 & 57506767355 \\ 0 & 99 & 11 & 27 & 46 & 95 & 62 & 88 & 71 & 27 \\ 0 & 0 & 11 & 0 & 1 & 4 & 2 & 4 & 0 & 10 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

## Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of +1. Certified value:  $D_x(e_3) = (0 \ 4 \ 1)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 4: character $b$ , column $e_1$

Prescribed character vector  $(0 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

**The auxiliary pair  $(t, I')$**

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $128889748995233 = 2557 \cdot 159499 \cdot 316031$ :

$$\begin{pmatrix} 128889748995233 & 40302964010751 & 107693741560834 & 37161965696106 & 118244994313278 & 30446808581562 & 95073237225944 & 110267200967319 & 53139033007834 & 102926644032108 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

**Sign check and certified value**

Residue prime: the stored prime ideal above  $\ell = 3$  with residue degree 2; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (0 \ 0 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

**Entry 5: character  $b$ , column  $e_2$**

Prescribed character vector  $(0 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

**The extension and its normalized generator**

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right) s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

**The auxiliary pair  $(t, I')$**

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $716420421540337 = 41 \cdot 3701 \cdot 4721337157$ :

$$\begin{pmatrix} 716420421540337 & 439915674276760 & 486577769761015 & 378703171675192 & 54620729612907 & 426757902429438 & 568713151003147 & 3560740732264 & 559693418986395 & 62380647010835 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

**Sign check and certified value**

Residue prime: the stored prime ideal above  $\ell = 3$  with residue degree 2; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (0 \ 0 \ 4)$  in the coordinates of  $\text{Cl}(K)/5$ .

**Entry 6: character  $b$ , column  $e_3$**

Prescribed character vector  $(0 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

**The extension and its normalized generator**

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right) s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $773848489785849 = 3^2 \cdot 131321 \cdot 654755641$ :

$$\begin{pmatrix} 257949496595283 & 249684560401265 & 92865342864585 & 88881048392873 & 94241407971941 & 206258727450266 & 229796109771003 & 95221440722517 & 81115071193515 & 61039465987735 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 2 & 2 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_3) = (0 \ 0 \ 3)$  in the coordinates of  $\text{Cl}(K)/5$ .

### Entry 7: character $c$ , column $e_1$

Prescribed character vector  $(0 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $1018586032545285 = 3^2 \cdot 5 \cdot 29^2 \cdot 43 \cdot 163 \cdot 3840017$ :

$$\begin{pmatrix} 11707885431555 & 8897715785590 & 8731229078173 & 8967021363412 & 4428454790729 & 6458710182158 & 7347614000588 & 5456952104405 & 6953438502534 & 9992979416085 \\ 0 & 29 & 26 & 19 & 8 & 18 & 16 & 9 & 24 & 3 \\ 0 & 0 & 3 & 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 324 factors with signed exponents of absolute value up to 2852864096420; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (3 \ 0 \ 0)$  in the coordinates of  $\text{Cl}(K)/5$ .

### Entry 8: character $c$ , column $e_2$

Prescribed character vector  $(0 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $53073656002475 = 5^2 \cdot 43 \cdot 71 \cdot 695363983$ :

$$\begin{pmatrix} 53073656002475 & 26349265848645 & 52254118441753 & 5162972146874 & 13650257953896 & 33232368743957 & 40154191976762 & 39969065660670 & 22965542274587 & 5470058977411 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 328 factors with signed exponents of absolute value up to 2125204679829; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (1 \ 0 \ 0)$  in the coordinates of  $\text{Cl}(K)/5$ .



## Entry 9: character $c$ , column $e_3$

Prescribed character vector  $(0 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $167516208537215 = 5 \cdot 187669 \cdot 178523047$ :

$$\begin{pmatrix} 167516208537215 & 63543303819781 & 40891670051371 & 163132407498721 & 105857733906588 & 59150729847957 & 105987569427054 & 79278162390118 & 158050259564401 & 112876643350814 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 1528099554654; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_3) = (0 \ 1 \ 0)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 10: character $a + b$ , column $e_1$

Prescribed character vector  $(1 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - \\ & 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682 \\ & 252975y + 33521132255139233 \end{aligned}$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$ .

## The pair $(a', J)$ representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

## The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $954790963349567 = 71^2 \cdot 189405071087$ :

$$\begin{pmatrix} 13447760047177 & 7301202517667 & 5099431484620 & 1726471814904 & 9561197763960 & 7957130067557 & 7711065935835 & 638629510102 & 6546143098024 & 6232563125703 \\ 0 & 71 & 47 & 49 & 10 & 25 & 26 & 63 & 3 & 50 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 1080 factors with signed exponents of absolute value up to 25310787346386697130381438695021134989048594; it is never expanded (its principal ideal and residues are evaluated factor by factor).

## Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (1 \ 4 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 11: character $a + b$ , column $e_2$

Prescribed character vector  $(1 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

**The auxiliary pair  $(t, I')$**

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $705111402369537 = 3^2 \cdot 71 \cdot 424093 \cdot 2601931$ :

$$\begin{pmatrix} 235037134123179 & 54781051842312 & 118746788783711 & 67082409614250 & 48814698257518 & 204769882102508 & 88737827282103 & 21600865656612 & 229950878499927 & 48241865615620 \\ 0 & 3 & 0 & 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 12167129602129086946573171804864191419967684; it is never expanded (its principal ideal and residues are evaluated factor by factor).

**Sign check and certified value**

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (0 \ 0 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

**Entry 12: character  $a + b$ , column  $e_3$**

Prescribed character vector  $(1 \ 1 \ 0)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

**The extension and its normalized generator**

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

**The auxiliary pair  $(t, I')$**

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $417818487826197 = 3^2 \cdot 46424276425133$ :

$$\begin{pmatrix} 139272829275399 & 88713202440596 & 92755567741182 & 90177769625912 & 130026083504969 & 8390862915365 & 34713384729554 & 17208681903498 & 64312609986576 & 50418878540397 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 2 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 11687433564430013207989635317620972294978905; it is never expanded (its principal ideal and residues are evaluated factor by factor).

**Sign check and certified value**

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_3) = (1 \ 4 \ 0)$  in the coordinates of  $\text{Cl}(K)/5$ .

**Entry 13: character  $a + c$ , column  $e_1$**

Prescribed character vector  $(1 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

**The extension and its normalized generator**

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

**The pair  $(a', J)$  representing the torsion class**

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a') J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $9464744484864 = 2^{13} \cdot 3^4 \cdot 257 \cdot 55501$ :

$$\begin{pmatrix} 684660336 & 222154512 & 650284596 & 683220186 & 175996950 & 234377421 & 130277844 & 520380619 & 111932259 & 323728374 \\ 0 & 24 & 16 & 8 & 22 & 8 & 0 & 1 & 15 & 20 \\ 0 & 0 & 4 & 2 & 0 & 2 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 6 & 0 & 0 & 4 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 43099987836648487482; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (1 \ 1 \ 4)$  in the coordinates of  $\text{Cl}(K)/5$ .

### Entry 14: character $a + c$ , column $e_2$

Prescribed character vector  $(1 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $25775647135776 = 2^5 \cdot 3^2 \cdot 4621 \cdot 19367837$ :

$$\begin{pmatrix} 1073985297324 & 5225662018 & 630948454836 & 1062010302135 & 354200203553 & 416677103064 & 535373255772 & 421369491208 & 285924777230 & 895865301630 \\ 0 & 2 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 12 & 11 & 8 & 7 & 7 & 0 & 9 & 9 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 40051445026885797606; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (3 \ 0 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

### Entry 15: character $a + c$ , column $e_3$

Prescribed character vector  $(1 \ 0 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $91146205970432 = 2^{14} \cdot 5563122923$ :

$$\begin{pmatrix} 356039867072 & 252209300936 & 66898099604 & 18048622616 & 229209357714 & 236761310544 & 2098126469 & 310097952939 & 319692912162 & 156875209885 \\ 0 & 8 & 4 & 0 & 3 & 4 & 4 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 3 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 & 6 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 437 factors with signed exponents of absolute value up to 30658864764070769427; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 7$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_3) = (4 \ 2 \ 1)$  in the coordinates of  $\text{Cl}(K)/5$ .

### Entry 16: character $b + c$ , column $e_1$

Prescribed character vector  $(0 \ 1 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $80312990617728 = 2^7 \cdot 3^4 \cdot 11 \cdot 101 \cdot 6972311$ :

$$\begin{pmatrix} 557729101512 & 349832153790 & 462355049748 & 274706643287 & 342513048316 & 365118894658 & 111611279999 & 355101108901 & 50239291891 & 23562060756 \\ 0 & 18 & 0 & 1 & 2 & 0 & 8 & 4 & 10 & 9 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 254 factors with signed exponents of absolute value up to 4061008623; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_1) = (3 \ 3 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 17: character $b + c$ , column $e_2$

Prescribed character vector  $(0 \ 1 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .

### The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

### The pair $(a', J)$ representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

### The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $461557383369984 = 2^8 \cdot 3^{10} \cdot 11^2 \cdot 252341$ :

$$\begin{pmatrix} 1798686648 & 666047448 & 607042272 & 655620738 & 108376403 & 37304866 & 299819624 & 970450028 & 1288038357 & 920414049 \\ 0 & 1782 & 1716 & 501 & 709 & 859 & 488 & 369 & 599 & 294 \\ 0 & 0 & 12 & 6 & 8 & 9 & 2 & 9 & 4 & 6 \\ 0 & 0 & 0 & 6 & 3 & 1 & 4 & 3 & 5 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 259 factors with signed exponents of absolute value up to 2003087558; it is never expanded (its principal ideal and residues are evaluated factor by factor).

### Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_2) = (1 \ 1 \ 4)$  in the coordinates of  $\text{Cl}(K)/5$ .

## Entry 18: character $b + c$ , column $e_3$

Prescribed character vector  $(0 \ 1 \ 1)$ , i.e. the values of  $x$  on the classes of the three stored generators of  $\text{Cl}(K)$ .



## The extension and its normalized generator

Relative polynomial of  $L_x$  over  $K$ :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of  $L_x$  over  $\mathbf{Q}$  (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism  $\sigma_x$  is determined by the image of the primitive element  $\theta$ : the coordinates of  $\sigma_x(\theta)$  in the integral basis of  $L_x$  are  $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$ .

## The pair $(a', J)$ representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$ . **Checked:**  $(a')J^5 = \mathcal{O}_K$ : pass.

## The auxiliary pair $(t, I')$

$I'$  is the ideal of  $L_x$  with the following HNF matrix in the integral basis of the absolute model; its absolute norm is  $457970075887104 = 2^9 \cdot 3^2 \cdot 11^3 \cdot 74670073$ :

$$\begin{pmatrix} 19712899272 & 7067514300 & 19703872672 & 12268240548 & 5325940829 & 78057772 & 2095614532 & 17306151784 & 6427559407 & 5934028045 \\ 0 & 132 & 44 & 76 & 66 & 44 & 56 & 33 & 120 & 126 \\ 0 & 0 & 22 & 0 & 0 & 8 & 0 & 21 & 12 & 6 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element  $t$  is stored in compact form as a product of 258 factors with signed exponents of absolute value up to 3995280885; it is never expanded (its principal ideal and residues are evaluated factor by factor).

## Sign check and certified value

Residue prime: the stored prime ideal above  $\ell = 5$  with residue degree 1; there the quotient  $N_x(t)/a'$  reduces to the residue of  $+1$ . Certified value:  $D_x(e_3) = (2 \ 3 \ 2)$  in the coordinates of  $\text{Cl}(K)/5$ .